

Generating Wikipedia Articles from Citations

Nino Reil, Andreas Walln fer, Zoltan Bek, Armin Lohse

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1 Problem Description

Wikipedia articles are usually written as concise, neutral, and well-structured summaries of knowledge that is grounded in external sources. In practice, the references of a Wikipedia article often point to books, news articles, reports, scientific papers, or official websites that support individual claims in the text. This project asks the question: *Can we reconstruct a plausible Wikipedia-style article if we are only given the cited source material, but not the original article itself?*

A central challenge is that the cited sources may differ strongly in tone, structure, and writing style, while the generated output should follow the neutral, professional, and objective style typical of Wikipedia articles. The system must therefore not only extract relevant information from the sources, but also transform it into a consistent encyclopedic form.

2 Dataset Description

We will build the dataset from existing Wikipedia articles and their references.

- **Dataset Source:** Wikipedia articles together with their cited external references.
- **Type of Text:** Encyclopedic article text, citation metadata, and source documents such as news articles, reports, official websites, and scientific abstracts.
- **Approximate Size:** About 420 Wikipedia articles.
- **Language:** English.
- **Public or Collected:** Collected from public sources.

To keep the task feasible and the source coverage reliable, we will only include Wikipedia articles that satisfy two conditions:

- At least 20 citations in total.
- At least 90% of these citations are accessible to us, for example because they are not paywalled or missing.

Each sample consists of one Wikipedia article and its citation set. The original article text is used only as a reference target for evaluation, not as input to the generation system.

This is mainly a generation task, so the dataset does not contain class labels in the standard classification sense. Instead, each sample has:

- **Input:** Cited source documents and citation metadata.
- **Target:** The original Wikipedia article text.

3 Methodology Approach

The dataset (420 articles and their citations) will be divided into

1. **Training:** 300
2. **Validation:** 60
3. **Test:** 60

The validation set will be used for model selection and parameter tuning, while the test set will be kept untouched for the final evaluation.

We plan to compare at least two approaches.

Preprocessing Steps:

- Extract citations from selected Wikipedia articles.
- Download or collect accessible referenced documents.
- Remove web noise such as navigation text or advertisements.
- Split sources into passages.
- Extract metadata such as title, date, publisher, and named entities.

Embedding Model: We will use a transformer-based embedding model to represent source passages and support retrieval of relevant evidence for generation.

In addition to selecting relevant content, the system must normalize stylistic differences between the cited sources and transform them into a uniform Wikipedia-like style.

Approach 1: Baseline. A simpler baseline will rank important passages from the cited sources and create a multi-document summary, for example using extractive summarization or a lightweight generation pipeline.

Approach 2: Advanced Model. A more advanced approach will use retrieval-augmented generation with a transformer model. Relevant passages will be retrieved for each article section or generation step, and the model will generate a Wikipedia-style article grounded in these passages.

If time permits, we may add a structured variant in which the system first predicts an outline or section plan and then generates the final text section by section.

4 Evaluation Metrics

The models will be evaluated automatically against the original Wikipedia article, which is not provided as input to the system.

Since exact wording may differ even in a good reconstruction, we will not rely on a single metric. We plan to compare the generated article with the original Wikipedia article using

1. **Title Similarity:** Cosine similarity between sentence-transformer embeddings.
2. **Lexical Content Similarity:** ROUGE-1, ROUGE-2, and ROUGE-L.
3. **Semantic Content Similarity:** BERTScore.
4. **Absolute Difference in Section Count**
5. **Article Length Ratio**

5 Expected Results

We expect the transformer-based retrieval-augmented approach to outperform the baseline because it can better capture context across multiple sources and generate more coherent text.

One of the main difficulties will be bridging the gap between heterogeneous source language and the neutral, professional, and objective wording expected in Wikipedia articles. In general, we expect the generated articles to recover the main facts and overall topic, but to differ from the original Wikipedia article in wording, structure, and level of detail.